Blockchain DLOC Sem VII

NMCPC62 : Cryptocurrency and Blockchain Development

Module - 3: Cryptocurrency Ecosystem and Wallets (6 Hours)

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Topics to be covered

1. Popular Cryptocurrencies: Bitcoin, Ethereum, Ripple, and Altcoins,
2. Cryptocurrency Mining: Mechanisms, Challenges, and Rewards,
3. Cryptocurrency Wallets: Hot vs. Cold Wallets,
4. Multi-Signature Wallets.
5. Securing Wallets: Threats, Recovery Methods, and Best Practices,
6. Cryptocurrency Exchanges: Types, Trading Mechanisms, and Risks,
7. Regulatory and Legal Frameworks for Cryptocurrencies Globally

1. Popular Cryptocurrencies - Bitcoin (BTC)

- Launched: 2009
- Founder: Satoshi Nakamoto
- Primary Purpose: Peer-to-peer digital currency

Key Features:

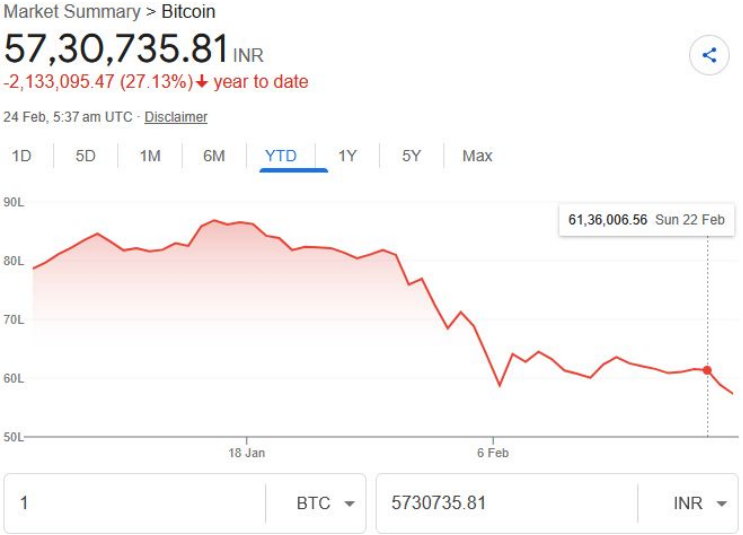
- First cryptocurrency (pioneer of blockchain technology)
- Fixed supply: 21 million BTC
- Proof of Work (PoW) consensus
- Highly decentralized
- Store of value ("Digital Gold")

Strengths:

- Most secure and decentralized
- Highest market capitalization
- Widely accepted

Limitations:

- Slower transaction speed (~7 TPS)
- Higher energy consumption



Courtesy : Himanshi Blog

1. Popular Cryptocurrencies - Ethereum (ETH)

- Launched: 2015
- Founder: Vitalik Buterin
- Primary Purpose: Smart contracts & decentralized applications (DApps)

Key Features:

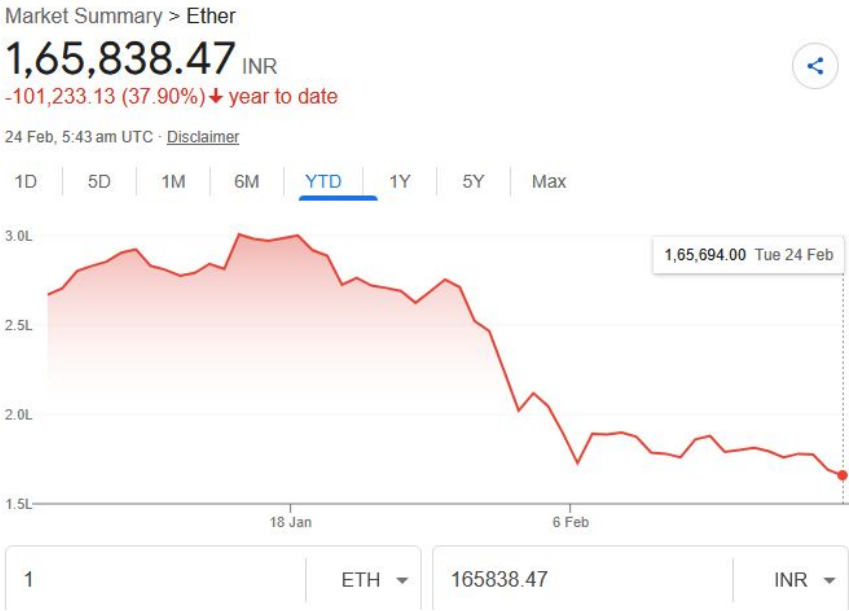
- Programmable blockchain
- Smart contract functionality
- Transitioned to Proof of Stake (PoS) (Ethereum 2.0)
- Native token: Ether (ETH)

Strengths:

- Foundation for DeFi, NFTs, Web3
- Large developer community
- Flexible programmable platform

Limitations:

- Gas fees can be high
- Scalability challenges (improving with Layer 2 solutions)



1. Popular Cryptocurrencies - Ripple (XRP)

- Launched: 2012
- Company Behind It: Ripple Labs
- Primary Purpose: Cross-border payments for financial institutions

Key Features:

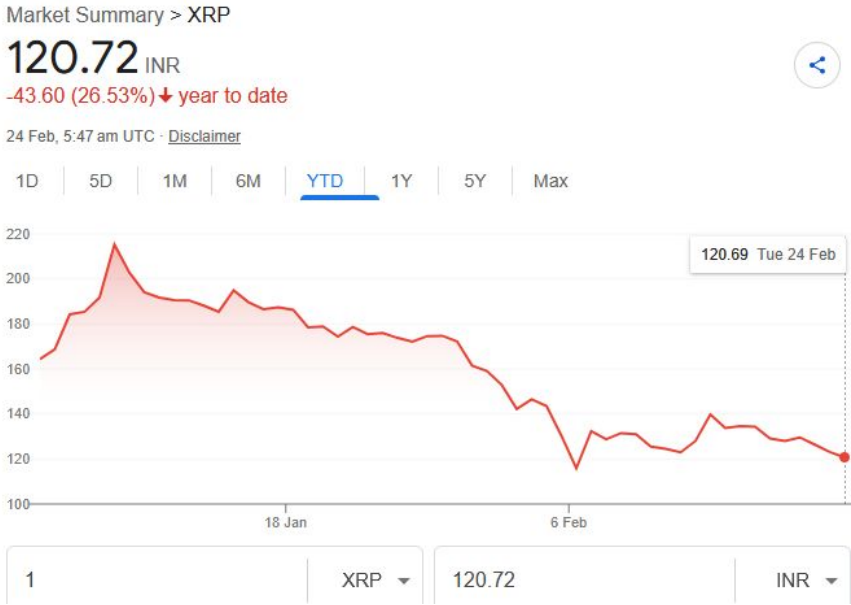
- Focused on banking & remittances
- Faster transactions (~3–5 seconds)
- Very low transaction fees
- Not fully decentralized

Strengths:

- Extremely fast settlement
- Energy efficient
- Strong banking partnerships

Limitations:

- More centralized than BTC/ETH
- Regulatory controversies (e.g., SEC case)



Courtesy : Himanshi Blog

1. Popular Cryptocurrencies - Altcoins

- Altcoin : Any cryptocurrency other than Bitcoin.
- Ex : Ethereum, Ripple, Litecoin, Cardano, Solana, Polkadot, etc…

Categories of Altcoins:

- Smart contract platforms
- Stablecoins
- Privacy coins
- Meme coins
- Governance tokens

Strengths:

- Innovation & experimentation
- Higher growth potential
- Specialized use cases

Risks:

- Higher volatility
- Some lack long-term sustainability
- Security risks in smaller projects

Courtesy : Binance

1. Popular Cryptocurrencies

Feature	Bitcoin	Ethereum	Ripple (XRP)	Altcoins
Launch Year	2009	2015	2012	Mostly post-2011
Founder	Satoshi Nakamoto	Vitalik Buterin	Ripple Labs	Various
Purpose	Digital currency	Smart contracts & DApps	Bank payments	Varies
Consensus	PoW	PoS	XRP Ledger Consensus	PoW/PoS/Other
Supply	Fixed (21M)	No fixed cap	Pre-mined	Varies
Speed	Slow	Moderate	Very Fast	Varies
Decentralization	Very High	High	Moderate	Varies

Courtesy : Himanshi Blog

1. Popular Cryptocurrencies

Feature	Binance Coin (BNB)	Solana (SOL)	Cardano (ADA)	XRP	Dogecoin (DOGE)
Launch Year	2017	2020	2017	2012	2013
Founder	Changpeng Zhao	Anatoly Yakovenko	Charles Hoskinson	Ripple Labs	Billy Markus & Jackson Palmer
Primary Purpose	Exchange utility & smart contracts	High-speed DApps	Research-driven smart contracts	Cross-borde r payments	Peer-to-peer payments / Meme coin
Consensus	PoS Authority (BSC)	PoH + PoS	Ouroboros PoS	XRP Ledger Consensus	PoW (Scrypt)
Supply	Limited (~200M, periodic burns)	Inflationary (capped growth model)	Fixed max (45B)	Pre-mined (100B)	Inflationary (no hard cap)

Courtesy : Himanshi Blog

1. Popular Cryptocurrencies

Feature	Binance Coin (BNB)	Solana (SOL)	Cardano (ADA)	XRP	Dogecoin (DOGE)
Speed (TPS approx.)	~2,000+	~65,000 (theoretical)	~250+	~1,500	~30
Decentralization	Moderate	Moderate	High	Moderate	Moderate–Low
Energy Usage	Low	Low	Very Low	Very Low	Moderate
Key Strength	Strong exchange ecosystem	Very high throughput	Academic & peer-reviewed development	Banking adoption	Strong community

Courtesy : Himanshi Blog

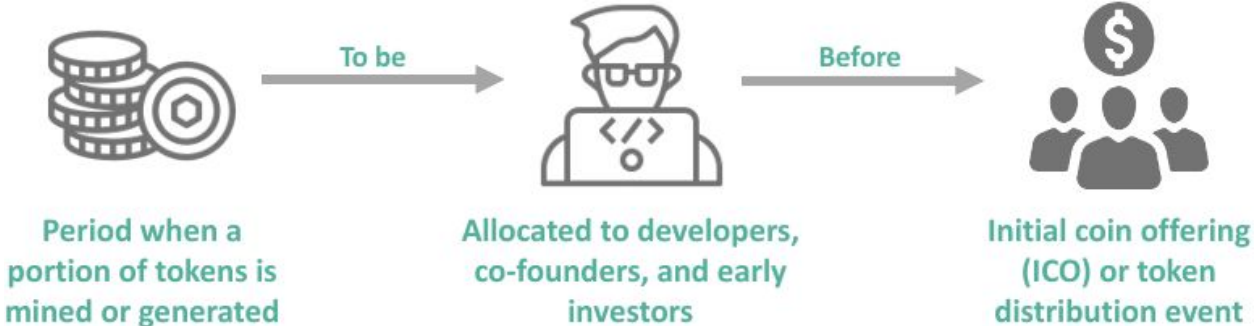
2. Cryptocurrency Mining: Mechanisms, Challenges, and Rewards

Pre-mined : cryptocurrency whose total or partial supply of coins/tokens was created (generated) before

- the blockchain was made public **or**
- before general participants could mine or validate blocks.

How Pre-mining Works ?

1. Developers create the entire supply (or a large portion).
2. Tokens are allocated to:
   - Founders
   - Development team
   - Foundation
   - Investors
   - Ecosystem incentives
3. Tokens may be:
   - Locked in escrow
   - Released periodically
   - Sold in public/private sales



Courtesy : Pre-Mining

2. Cryptocurrency Mining: Mechanisms, Challenges, and Rewards

Feature	Pre-mined	Mined (e.g., Bitcoin)
Coin Creation	Created before launch	Created gradually
Mining Required	No	Yes
Supply Release	Controlled by project	Algorithm-based
Decentralization	Often lower	Usually higher
Example	XRP	Bitcoin

Courtesy : Himanshi Blog

2. Cryptocurrency Mining: Mechanisms, Challenges, and Rewards

1. Bitcoin (BTC)

Mining Mechanism

- Uses Proof of Work (PoW) consensus.
- Miners solve complex cryptographic hash puzzles (SHA-256).
- First miner to solve the puzzle adds the block to the blockchain.
- Requires specialized hardware (ASIC miners).

Process

1. Transaction verification
2. Block formation
3. Hash computation
4. Block validation
5. Reward distribution

Rewards

- Block reward (currently reduced periodically via halving)
- Transaction fees
- Incentive for securing the network

Challenges

- Extremely high energy consumption
- Expensive hardware investment
- Mining difficulty increases over time
- Environmental concerns

2. Cryptocurrency Mining: Mechanisms, Challenges, and Rewards

2. Ethereum (ETH)

Mining Mechanism (Before 2022)

- Used Proof of Work (Ethash algorithm)
- GPU-based mining

Current Mechanism (Post "Merge")

Ethereum transitioned to Proof of Stake (PoS) in 2022.

- Validators stake 32 ETH
- Blocks are validated based on stake and randomness
- No traditional mining

Rewards

- Staking rewards
- Transaction (gas) fees
- MEV (Maximal Extractable Value)

Challenges

- Requires 32 ETH to run independent validator
- Slashing penalties for malicious behavior
- Network congestion impacts rewards

2. Cryptocurrency Mining: Mechanisms, Challenges, and Rewards

No Mining Mechanism

Unlike Bitcoin and early Ethereum:

- XRP is not mined
- All tokens were pre-mined at launch
- Uses Ripple Protocol Consensus Algorithm (RPCA)

Validation Process

- Trusted validator nodes reach consensus
- No block rewards
- Minimal transaction fees (burned)

3. Ripple (XRP)

Developed by Ripple Labs.

Rewards

- No mining rewards
- Incentives based on network participation & institutional adoption

Challenges

- More centralized validator structure
- Regulatory scrutiny

2. Cryptocurrency Mining: Mechanisms, Challenges, and Rewards

Feature	Bitcoin	Ethereum	Ripple (XRP)	Altcoins
Mining Type	PoW	PoS (after Merge)	No Mining	PoW / PoS / Hybrid
Hardware Needed	ASIC	None (Staking)	None	GPU / Stake
Energy Use	Very High	Low	Very Low	Varies
Reward Type	Block + Fees	Staking + Fees	None	Block / Staking
Entry Barrier	High	Medium–High	N/A	Varies

Courtesy : Himanshi Blog

2. Cryptocurrency Mining: Mechanisms, Challenges, and Rewards

Proof of Authority / PoS Hybrid – BNB

- Uses Proof of Staked Authority.
- Limited validator set → faster blocks.
- More centralized compared to Cardano.

Proof of History + PoS – Solana

- Uses cryptographic time-stamping.
- Very high throughput.
- Validator hardware requirements are relatively high.

Pure Proof of Stake – Cardano

- Ouroboros protocol (peer-reviewed).
- Stake pools validate blocks.
- Highly energy efficient.

Consensus without Mining – XRP

- No mining or staking.
- Validators reach agreement via Ripple consensus.
- Minimal energy usage.

Proof of Work – Dogecoin

- Scrypt-based PoW (similar to Litecoin).
- Miners compete to solve cryptographic puzzles.
- Inflationary reward model.

2. Cryptocurrency Mining: Mechanisms, Challenges, and Rewards

Feature	BNB	Solana	Cardano	XRP	Dogecoin
Mining Type	No mining (PoSA)	No mining (PoH + PoS)	No mining (PoS – Ouroboros [peer-reviewed]	No mining (Consensus Protocol)	Proof of Work (Scrypt)
Block Validation	Staked validators	Staked validators + time-stamping	Stake pools validate blocks	Trusted validator nodes	Miners solve hash puzzles
Hardware Required	None	None	None	None	GPU / ASIC (Scrypt)
Energy Consumption	Low	Low	Very Low	Very Low	Moderate

Courtesy : Himanshi Blog

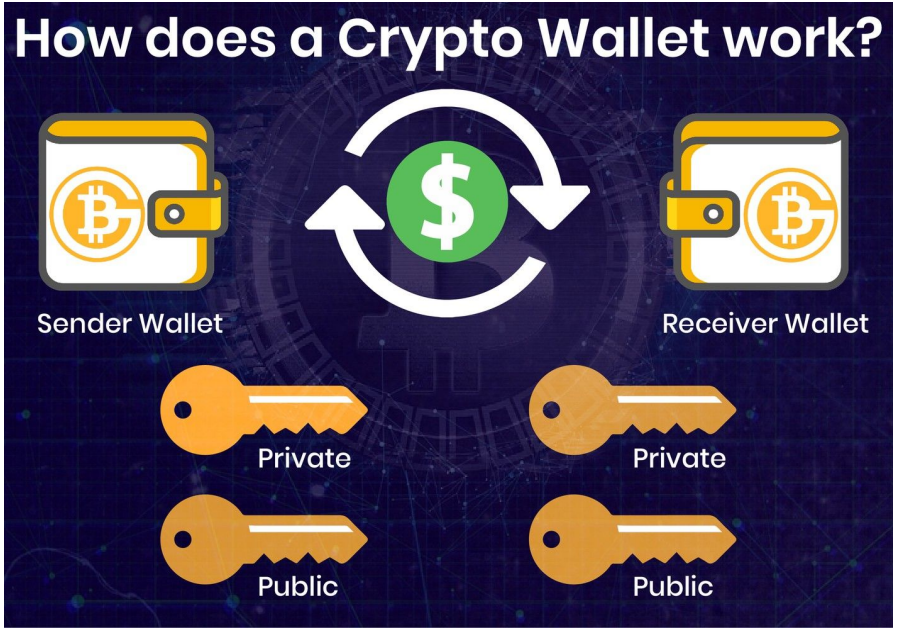
2. Cryptocurrency Mining: Mechanisms, Challenges, and Rewards

Feature	BNB	Solana	Cardano	XRP	Dogecoin
Reward Type	Staking rewards + ecosystem incentives	Staking rewards + fees	Staking rewards + fees	No block rewards	Block rewards + transaction fees
Token Creation	Pre-issued with burns	Inflationary	Fixed max (45B)	Fully pre-mined	Inflationary
Entry Barrier	Need BNB stake	Stake SOL	Delegate or run stake pool	Validator participation limited	Hardware + electricity cost
51% Attack Risk	Low (limited validators)	Low–Moderate	Low	Very Low	Moderate (PoW-based)

Courtesy : Himanshi Blog

3. Cryptocurrency Wallets: Hot vs. Cold Wallets

- A cryptocurrency wallet is a **software program that stores your digital money**.
- **store your private keys** - the passwords that give you access to your cryptocurrencies
- Keeps the **crypto safe and accessible.**
- **allow the user to send, receive, and spend cryptocurrencies** like Bitcoin and Ethereum.



Courtesy : Coinbase,
Benzinga
Blockchain Simplified

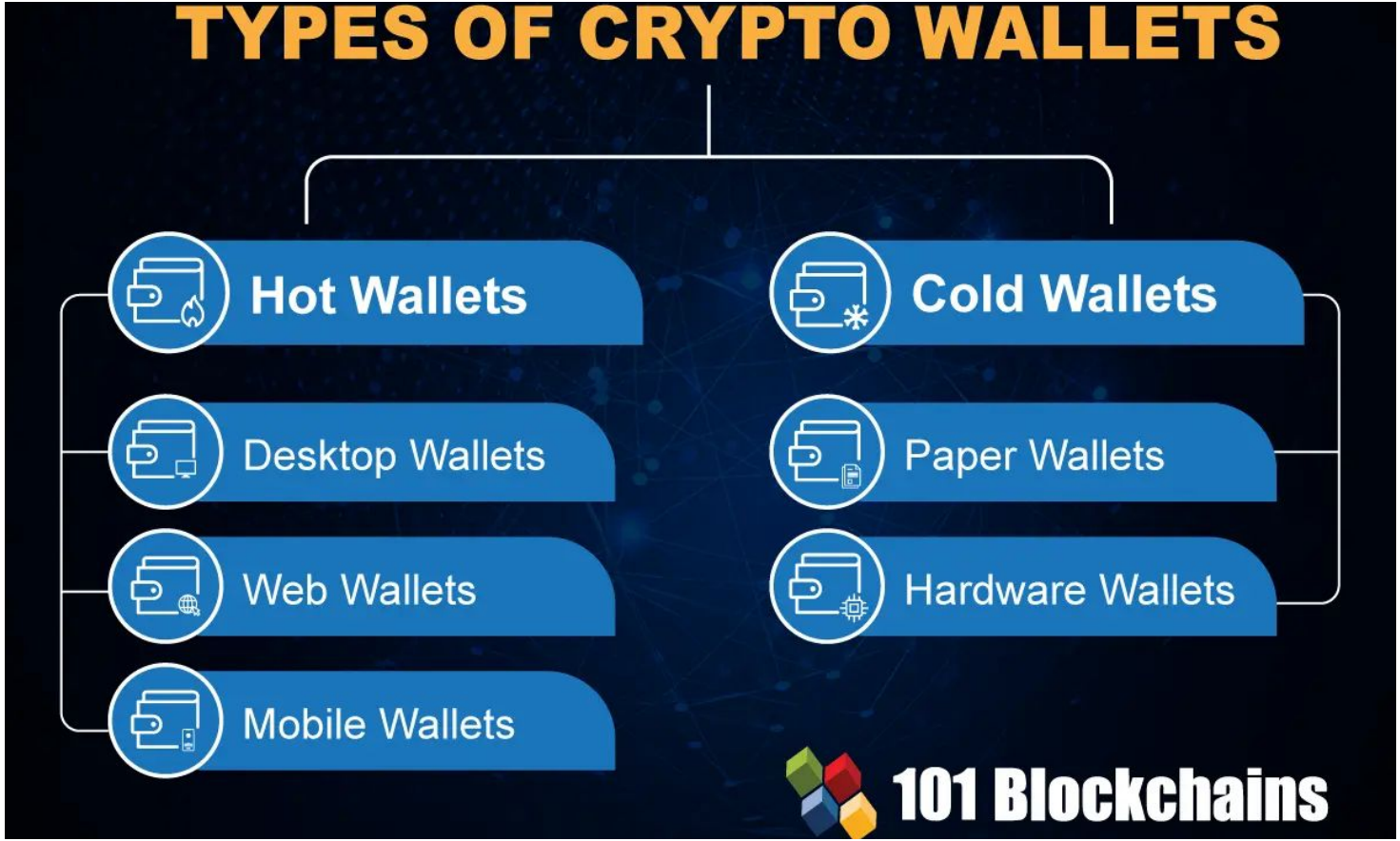
3. Cryptocurrency Wallets: Hot vs. Cold Wallets

Why are crypto wallets important?

- Normal wallet - holds actual cash,
- Crypto wallets
  - **don't store your crypto** instead store **private keys**
    - used to access live holdings on the blockchain
    - prove the ownership of your digital money
    - Allows to perform transactions.
    - Losing private keys ⇒ losing access to your money.
  - That's why it's i**mportant to keep the hardware wallet safe, or use a trusted wallet provider like Coinbase.**

Courtesy : Coinbase

3. Cryptocurrency Wallets: Hot vs. Cold Wallets



Courtesy : 101 Blockchains

3. Cryptocurrency Wallets: Hot vs. Cold Wallets

HOT Wallets - Online Wallets

- also known as a **web wallet,**
- allows you **access to your cryptos via the Internet.**
- The online **wallet provider stores your crypto's private key on their server.**
- The **online wallet service website**
  - **send you the crypto code**,
  - will save your keys and
  - will give you the ability to access your keys.
- Advantages
  - **Enable fast transactions**.
  - May be **able to manage multiple cryptocurrencies.**
  - Conveniently **use on the go and for active trading**.
- Disadvantages:
  - **Risk of online security** such as hacks and scams.
  - **Risk of personal protection** such as computer viruses.
  - A **third-party is storing your cryptos**, not you.

Courtesy : Coinbase, Benzinga

3. Cryptocurrency Wallets: Hot vs. Cold Wallets

HOT Wallets - Mobile Wallets

- **available on cell phone through an app**.
- Just **like Apple Pay / Google Pay / PayTM**, you can use mobile wallets when shopping in physical stores as cryptocurrencies become more acceptable.
- Online wallets offer mobile versions as well.
- Advantages
  - Can be **safer than online wallets.**
  - Conveniently **use on the go.**
  - Offer **additional features such as QR code scanning.**
- Disadvantages:
  - **Risk of losing your crypto assets** if your phone is lost or damaged.
  - **Risk of mobile viruses and malware**.

Courtesy : Coinbase, Benzinga

3. Cryptocurrency Wallets: Hot vs. Cold Wallets

HOT Wallets - Desktop Wallet

- Another choice that can be **safer than online wallets**.
- *One can download your desktop wallet and install it on your computer*.
- **Are safe if the computer is not**, or even better, has never been **connected to the Internet.**
- If a **desktop computer has never been connected to the Internet**, ⇒ a **cold wallet.**
- Advantages of desktop wallets include:
  - A **convenient choice** for those who trade cryptos from their computers.
  - **Private keys are not stored on a third-party server**.
  - **If the computer is not connected to the Internet, it can be extremely safe.**
- Disadvantages of desktop wallets include:
  - **Harder to use your crypto-assets on the go.**
  - **If connected to the Internet, it turns into a less secure hot wallet.**
  - If **you don't backup your computer and it dies, you lose your cryptos**.

Courtesy : Coinbase, Benzinga

3. Cryptocurrency Wallets: Hot vs. Cold Wallets

COLD Wallets - Hardware Wallets

- Arguably be one of the **safest types of cryptocurrency wallets** out there.
- They **store your private keys on a device like a USB drive**.
- You are **still able to make online transactions.**
- **As they are offline most of the time, ⇒ a cold wallet.**
- Advantages of hardware wallets include:
  - One of the **safest crypto wallet options.**
  - **Great for storing large amounts of cryptocurrencies** that you don't want to use on a day-to-day basis.
- Disadvantages of hardware wallets include:
  - **Most expensive type of wallet**.
  - **Not as user-friendly** especially for beginners, but an absolute must for large crypto amounts.

Courtesy : Coinbase, Benzinga

3. Cryptocurrency Wallets: Hot vs. Cold Wallets

COLD Wallets - Paper Wallets

- A **super cold crypto wallet.**
- To use, one **need to print out your private and public keys**.
- **Send funds by transferring the money to wallet's public address**
- **Withdraw or send your currencies by entering your private keys** / **Scan the QR code** on the paper wallet.
- Advantages of paper wallets include:
  - Essentially **hacker-proof**.
  - **Not stored on a computer or mobile**.
  - **Not stored on a third-party server.**
- Disadvantages of paper wallets include:
  - **Not user-friendly** for none-geeks.
  - **Harder to use** for day-to-day transactions.
  - They **can catch fire**.

Courtesy : Coinbase, Benzinga

3. Cryptocurrency Wallets: Hot vs. Cold Wallets

	Hot wallet	Cold wallet
Price	These are usually free, and some pay interest on stored crypto.	These require the purchase of an external device, around $50 to $250.
Better for	Hot wallets are convenient to access and use for trading.	Cold wallets are better suited for long-term storage.
Maximum number of cryptos	Hot wallets can store anywhere from one to tens of thousands of cryptocurrencies.	Cold wallets store anywhere from 1,000 to tens of thousands.
Cybersecurity	Average. Because they are connected to the internet, they could potentially be vulnerable to hacking.	Excellent. They can't be accessed online, but they require security measures to keep them from getting damaged, lost or stolen.
Loss protection	Good. Most have recovery and backup options and can be accessed from multiple devices.	Average. Most have recovery and backup options for a lost password, but not for a lost device.
Ease of transfer to exchanges	Excellent. Hot wallets are easily accessible as the wallet is already internet-connected.	Average. Cold wallets require an extra step to connect online through USB, Wi-Fi or QR code.

Courtesy : NerdWallet

3. Cryptocurrency Wallets: Hot vs. Cold Wallets

Summary of Cryptocurrency Wallets

	Software	Online	Hardware	Paper
Pros	User-controlled Security For POS coins, allows minting	High convenience, Accessible from any browser without needing to download the Blockchain	Protects user's private keys, which are stored on the device Can be recovered with PIN and seed	Extremely secure. Can't be hacked using digital means Great for long term storage
Cons	Must download entire Blockchain for each type of coin/token	Susceptible to key logging hacks Can't stake POS coins Unknown level of security	Doesn't support all coins/tokens Can't stake POS coins	Inconvenient to use for transactions
Examples	Electrum, Armory	Blockchain, MyEtherWallet	Ledger, Trezor	

Courtesy : Coinbase, Benzinga

3. Cryptocurrency Wallets: Hot vs. Cold Wallets

3 Common cryptocurrency wallets

TYPES OF STORAGE	Hot Wallet	Cold Wallet	Custodial Wallet
HOW IT WORKS	Web-based. It is always connected to the internet and crypto network.	Offline. It is not connected to the crypto network or internet.	A third party stores your crypto for you.
WHERE IS THE PRIVATE KEY STORED	Online	Offline	Could be either online or offline.
BENEFITS	Convenience, ease of use.	Highest level of security.	Simplest option, most convenient.
DISADVANTAGES	Vulnerable to online attacks.	Cost of device, inconvenience for day-to-day transactions.	You need to trust the third party.

Courtesy : DBS Treasures

3. Cryptocurrency Wallets: Custodial Vs Non-Custodial Wallets

Custodial Wallets vs Non-Custodial Wallets

Custodial Wallet	Non-Custodial Wallet
Custodian or Third party has control over funds	Users have control of their funds
Not very secure as keys are with a third party	Extremely secure as users have their private keys
Accounts can be restored in case of lost keys	Funds cannot be recovered in case of lost keys
Transaction fees are comparatively lesser	Users bear the complete transaction fees
Slower withdrawals due to KYC/AML checks	Withdrawals are comparatively faster

Courtesy : Blockchain Simplified

3. Cryptocurrency Wallets: Custodial Vs Non-Custodial Wallets

Courtesy : Blockchain Simplified

3. Cryptocurrency Wallets: Custodial Vs Non-Custodial Wallets

3. Cryptocurrency Wallets: Hardware Vs Software wallets

	SafePal Hardware wallet	SafePal Software Wallet
Private Key Storage Location	Financial grade EAL5 Chip+ secure element	Users' Cellphone
Wallet Type	Decentralized wallet	Decentralized wallet
Access to internet	No	Yes
Management Tool	SafePal APP	SafePal APP
Import method	Mnemonic phrase	Private Key Mnemonic Phrase Keystore Observation Mode
Passphrase	Supported	Supported
Assets	All of the SafePal supported assets (xx chains and 30,000+ tokens)	All of the SafePal supported assets (xx chains and 30,000+ tokens)

Courtesy : SafePal

3. Cryptocurrency Wallets: Deterministic Vs Non-deterministic wallets

Classification of Wallets based on dependency of the keys

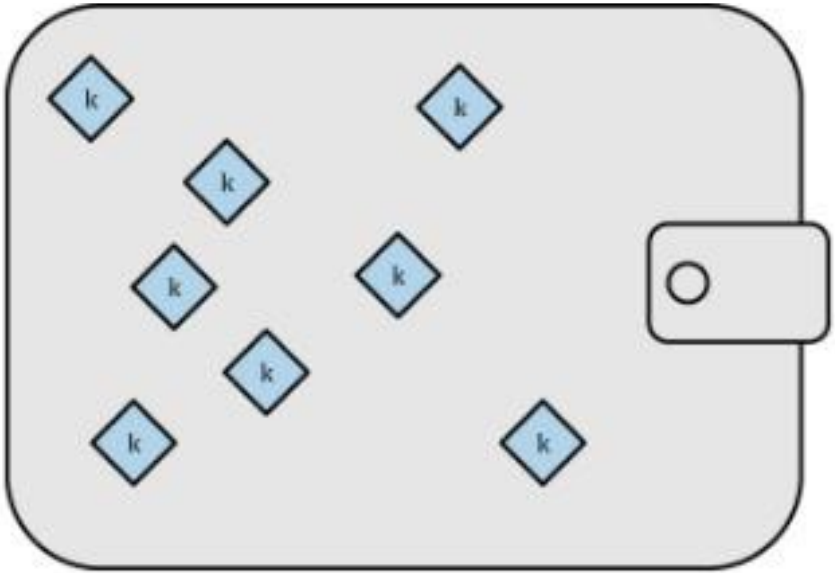
- **Non-deterministic wallet,**
  - each key is independently generated from a random number.
  - also known as a **JBOK "Just a Bunch Of Keys."**
- **Deterministic wallet**,
  - **All the keys are derived from a single master key ⇒ Seed.**
  - **All the keys** in this wallet **are related to each other and can be generated again if one has the original seed.**
  - The **most commonly used derivation method** uses a tree-like structure and is known as a **Hierarchical deterministic or HD wallet.**

Courtesy : Cointelegraph

3. Cryptocurrency Wallets: Deterministic Vs Non-deterministic wallets

Disadvantages of Nondeterministic (Random) Wallets

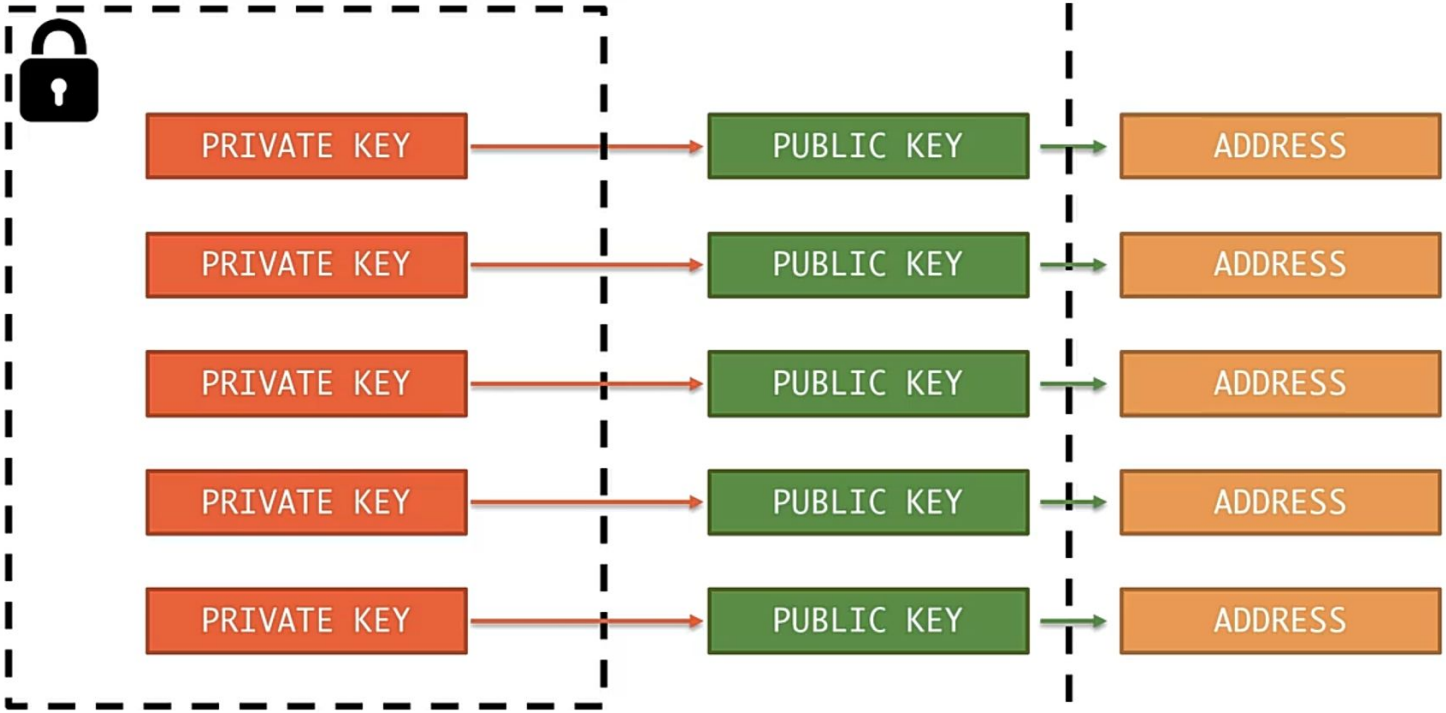
- Random keys generated needs to be backed up frequently.
- Address reuse reduces privacy by associating multiple transactions and addresses with each other.



Courtesy : Cointelegraph

3. Cryptocurrency Wallets: Deterministic Vs Non-deterministic wallets

Non-Deterministic Wallets



3. Cryptocurrency Wallets: Deterministic Vs Non-deterministic wallets

Deterministic (Seeded) Wallets

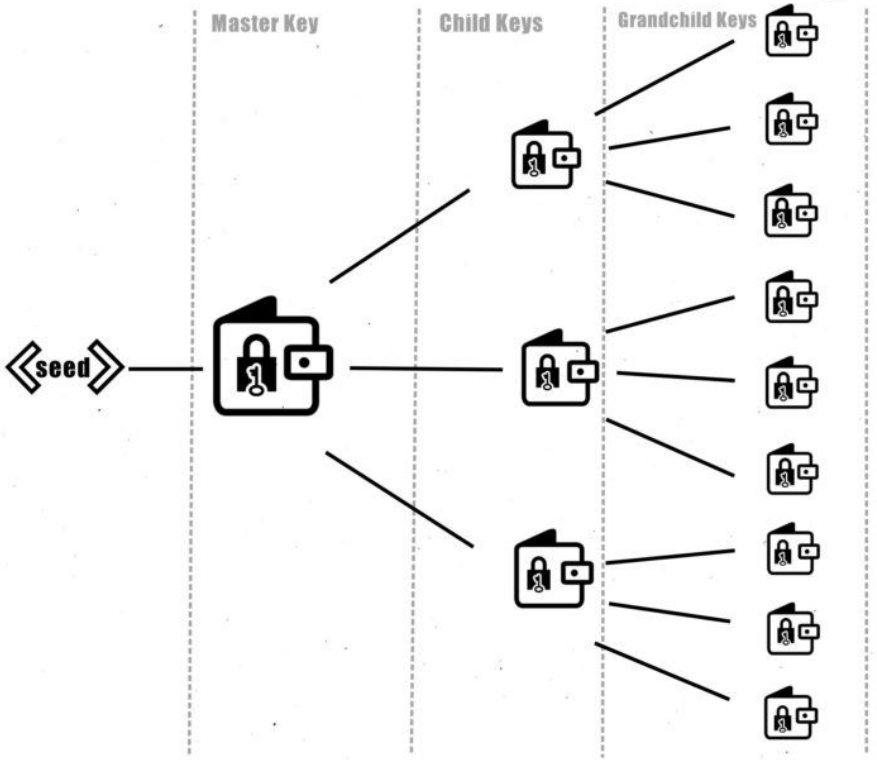
- contain **private keys that are all derived from a seed**, (one-way hash)
- Seed is a **randomly generated number**
  - combined with other data, an index number/ “chain code” to derive the private keys.
  - the seed is sufficient
    - to **recover all the derived keys,**
    - ensures **a single backup at creation time is sufficient.**
- wallet export / import, **allowing for easy migration of all the user’s keys between different wallet implementations.**

Courtesy : Cointelegraph

3. Cryptocurrency Wallets: Deterministic Vs Non-deterministic wallets

Hierarchical Deterministic Wallets (HD Wallets)

- Most advanced form of deterministic wallets
- defined by the **BIP-32** standard.
- contain keys **derived in a tree structure**, to an **infinite depth**
- Tree structure can be used to **express additional organizational meaning**,
- Eg: Branches of keys can also be used in corporate settings, allocating different branches to departments, subsidiaries, specific functions, or accounting categories.



Courtesy : Coingossip

3. Cryptocurrency Wallets: Deterministic Vs Non-deterministic wallets

Advantages of HD Wallets.

1. **No need to backup all your keys**, whenever you make transactions, **just backup the Seed praise securely.**
2. **Privacy is well guarded** using the hierarchical structure mechanism.
3. **Store multiple unique currencies within the same HD wallet**
4. **n-Number of public addresses can be generated to prevent re-use of addresses**.
5. **Backup of a private key from the hierarchical tree,** still your other addresses will remain safe.
6. **Move millions of dollars across borders without using a hardware wallet or mobile wallet app.**
7. **Users can create a sequence of public keys without having access to the corresponding private keys.** Thus **HD wallets to be used on an insecure server**
8. The **public keys do not need to be preloaded or derived in advance,**

Courtesy : Coingossip

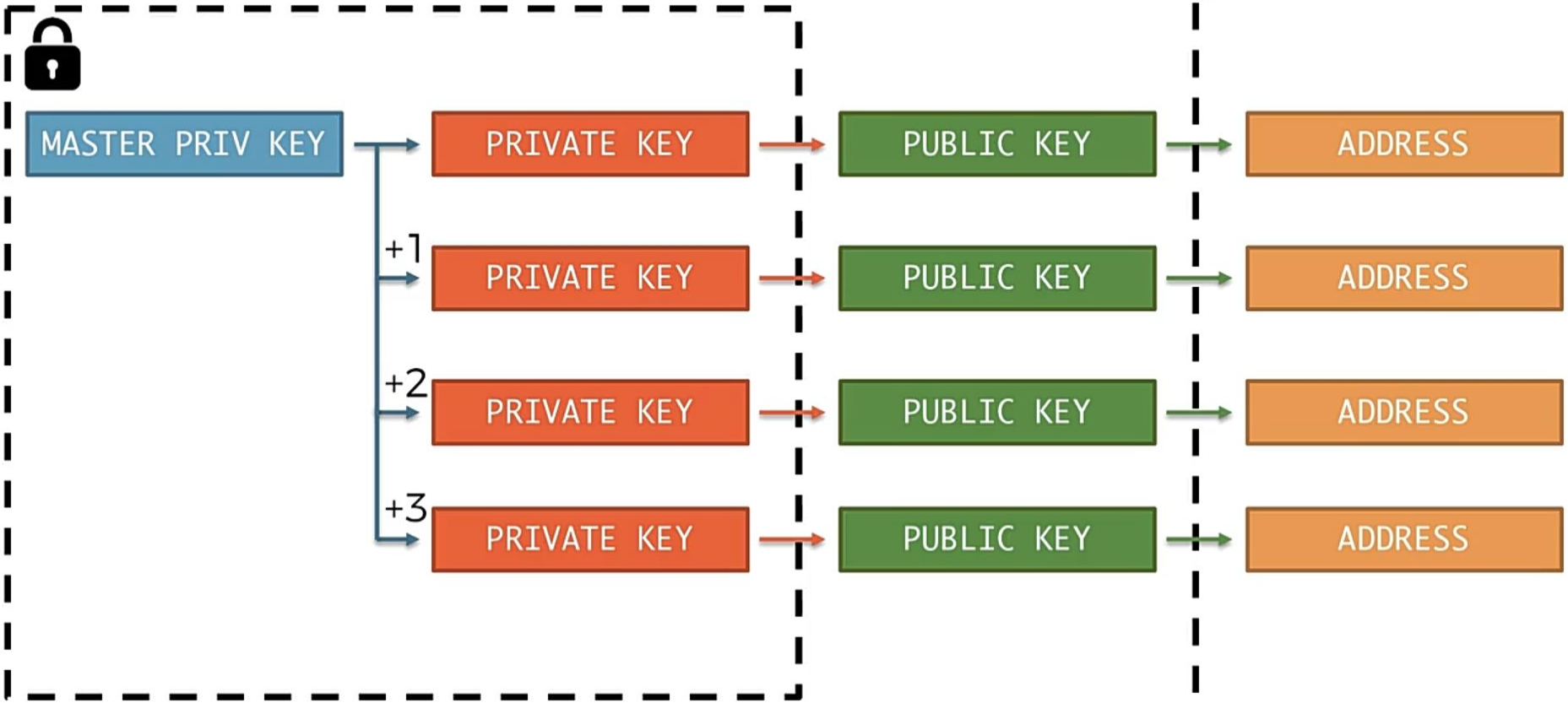
3. Cryptocurrency Wallets: Deterministic Vs Non-deterministic wallets

Disadvantages of HD wallets.

1. **Confuse new users**, as your **receiving address changes every time.**
2. If you chose **hard to memorizable seed praises then, it is difficult to backup all your key pairs** when the wallet is stolen or lost. **Chose memorizable key praise.**
3. you can **securely hand out child keys with no risk to the parent key,** and you **can hand out master public keys with no risk to the master private key**, you **cannot do both at the same time**.

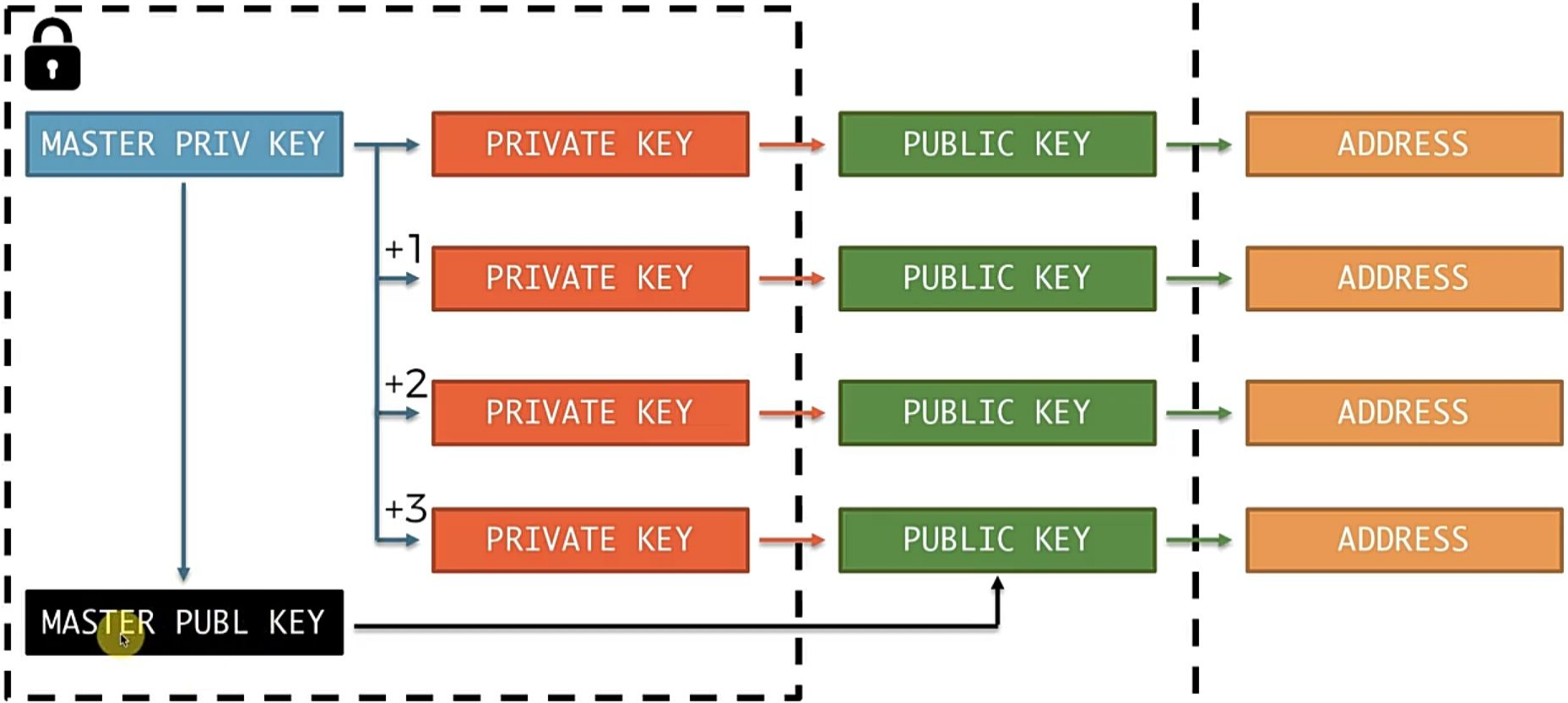
Courtesy : Coingossip

3. Cryptocurrency Wallets: Hierarchical Deterministic (HD) Wallets



Courtesy : Super Data Science

3. Cryptocurrency Wallets: Hierarchical Deterministic (HD) Wallets



3. Cryptocurrency Wallets: Hierarchical Deterministic (HD) Wallets

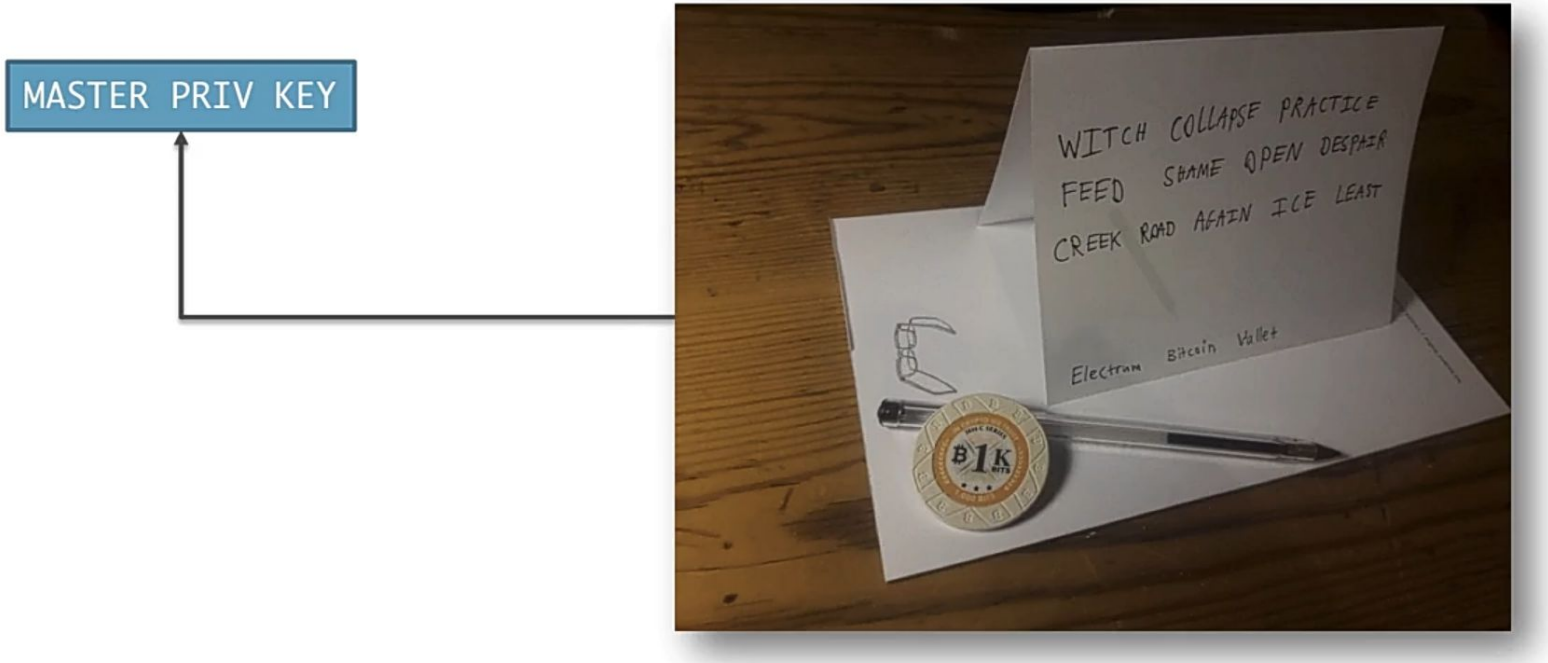


Image source: https://en.bitcoin.it/wiki/

3. Cryptocurrency Wallets:

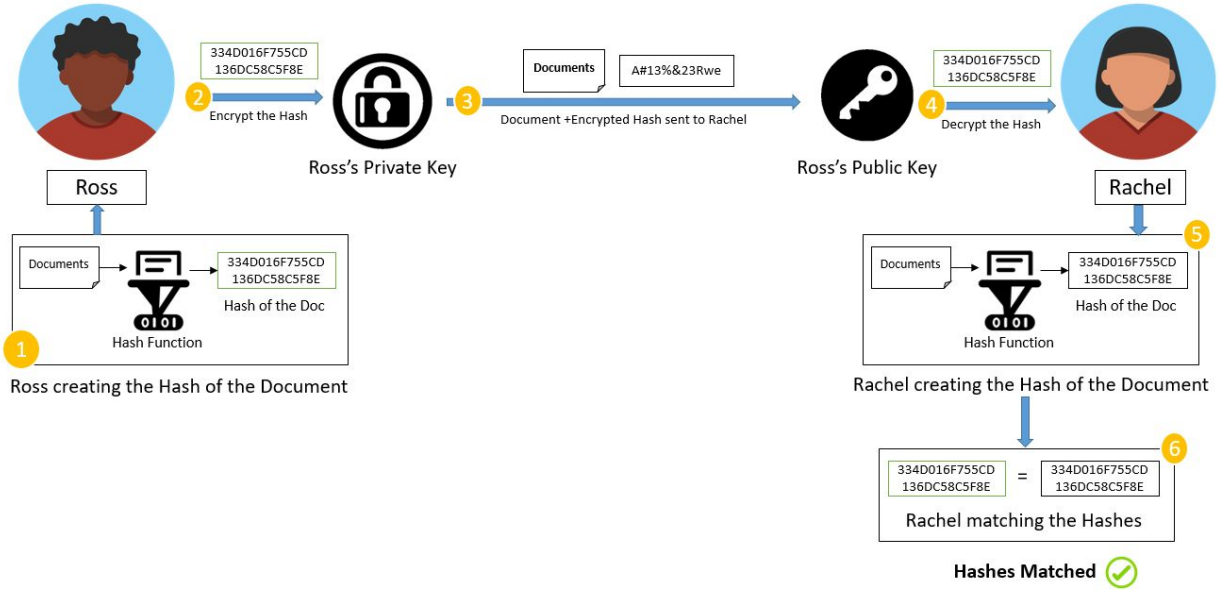
Wallet Best Practices

- certain common industry standards have emerged that **make bitcoin wallets broadly interoperable, easy to use, secure, and flexible.**
- These common standards are:
  - **Mnemonic code words**, based on **BIP-39**
  - **HD wallets**, based on **BIP-32**
  - **Multipurpose HD wallet structure**, based on **BIP-43**
  - **Multicurrency and multiaccount wallets**, based on **BIP-44**
- Eg : Breadwallet, Copay, Multibit HD, and Mycelium (software wallets)
- Eg : Hardware : Keepkey, Ledger, and Trezor.

4. Multi-Signature Wallets

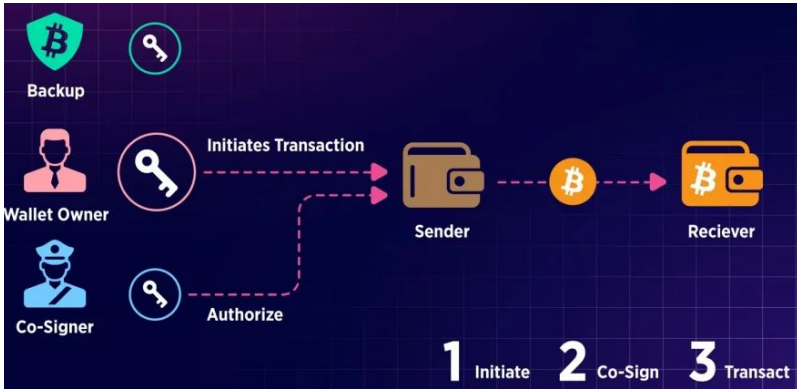
What is a Signature in Blockchain?

- Digital signature = cryptographic proof of ownership
- Generated using a private key
- Verifiable via corresponding public key
- Ensures:
  - Authentication
  - Integrity
  - Non-repudiation



4. Multi-Signature Wallets

- requires multiple private keys to authorize a transaction
- Improves security by distributing control
- Common in enterprise, DAO, and joint accounts
- Used in blockchain networks like Bitcoin and Ethereum



Why Use Multisig Wallets?

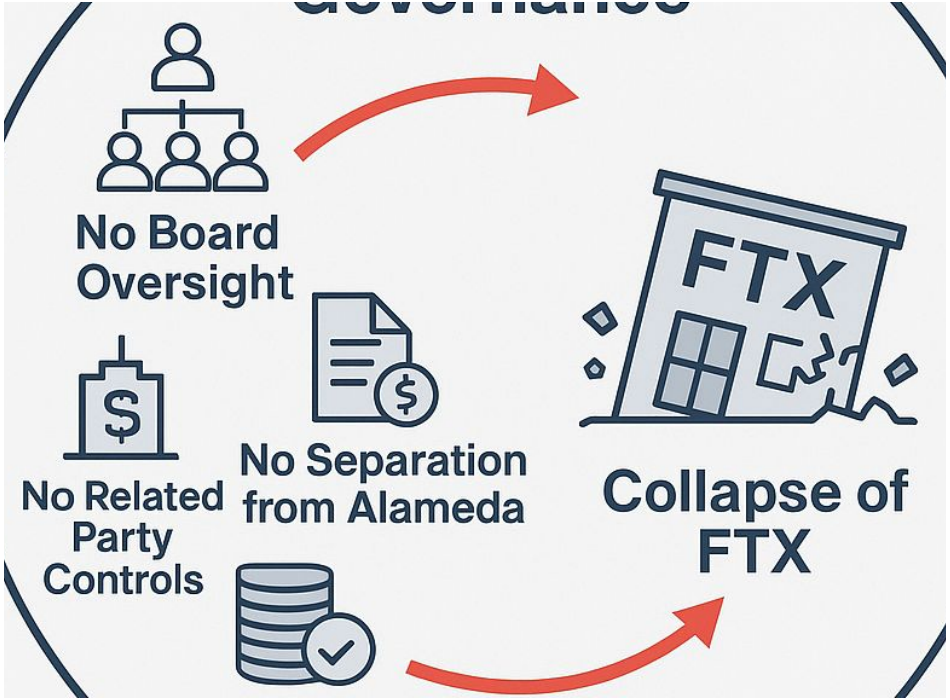
- Enhanced Security : Single key compromise does NOT drain funds
- Shared Control : Ideal for partnerships & organizations
- Corporate Treasury Management : Prevents insider fraud
- DAO Governance : Community-based fund control

Courtesy : CoinDesk

4. Multi-Signature Wallets

Real-World Importance :

- Eg. FTX collapse (2022) : (Multisig treasury management became a standard security practice in Web3 project

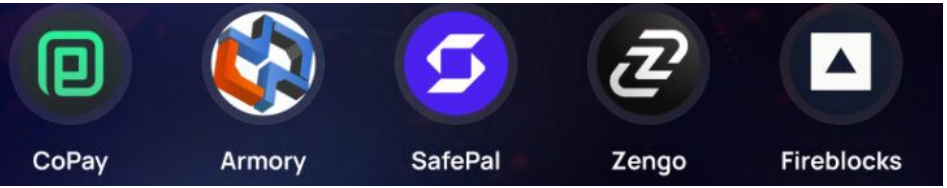


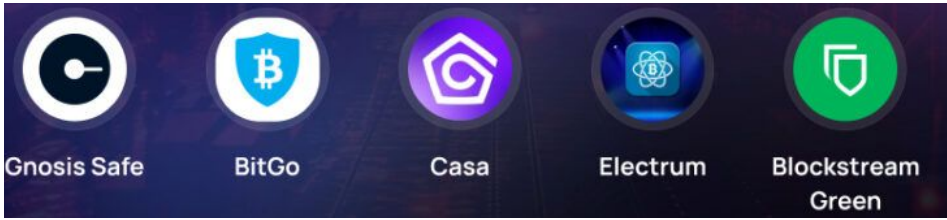
Courtesy : LinkedIn Post

4. Multi-Signature Wallets

Top Multisig Wallets & Platforms in 2026

- Electrum (Bitcoin) — free, open-source, very mature 2-of-3 & 3-of-5 support
- Casa — user-friendly 2-of-3 & 3-of-5 service (holds one key for you, very popular)
- Unchained Capital / Unchained Vault — collaborative custody 2-of-3 & 3-of-5
- Nunchuk — excellent mobile + desktop multisig (supports many chains)
- Sparrow Wallet — advanced Bitcoin multisig with strong privacy features
- SafePal (formerly Gnosis Safe) — Ethereum & EVM chains, dominant for DAOs
- BitGo — institutional-grade multisig (custodial & self-custody options)
- Fireblocks — enterprise / high-net-worth multisig



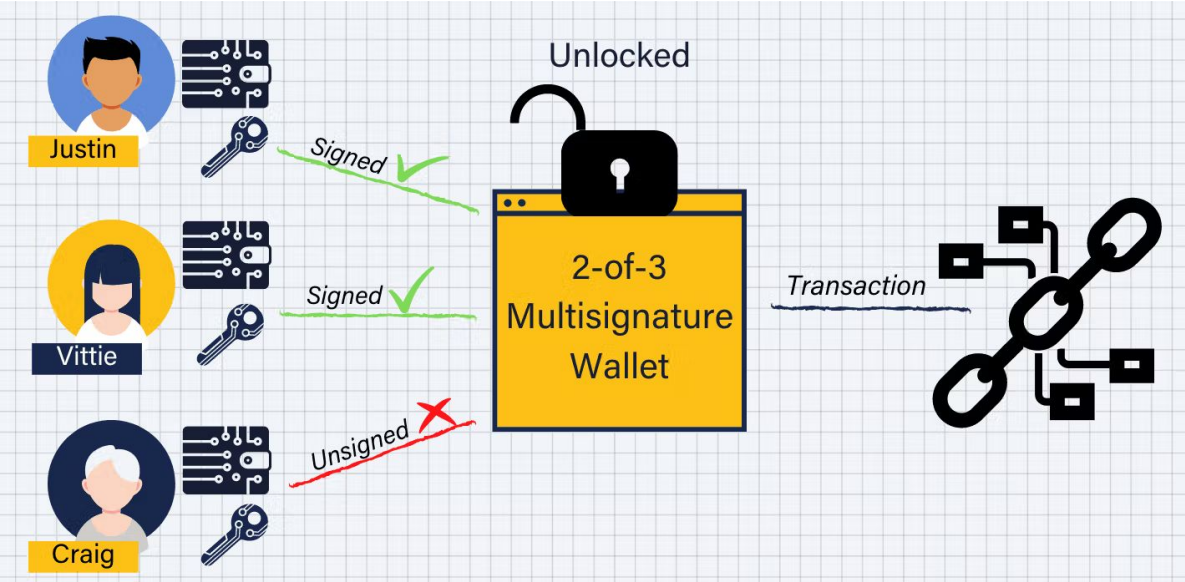


Courtesy : Calibriant

4. Multi-Signature Wallets

Types of Multisig Configurations

- 2-of-2 (Maximum security, no redundancy)
- 2-of-3 (Most common model)
- 3-of-5 (Enterprise setup)
- N-of-N (All parties must approve)



Courtesy : CoinDesk

4. Multi-Signature Wallets

How Multisig Works (Bitcoin Example)

- Uses Pay-to-Script-Hash (P2SH)
- Script defines:
  - Required number of signatures
  - Public keys allowed to sign
- Transaction valid only when conditions met

Multisig in Ethereum

- Implemented via smart contracts
- Wallet logic coded into contract
- Popular multisig wallet: Safe (formerly Gnosis Safe)

Use Cases

- Startup crypto treasury
- Investment funds
- DAO treasury management
- Escrow services
- Family inheritance planning

Courtesy : CryptoNews

4. Multi-Signature Wallets

Advantages of Multisig Wallets

✔ Reduced single point of failure

✔ Protection against theft

✔ Transparent transaction approval

✔ Better accountability

Disadvantages / Limitations

✘ Complex setup

✘ Coordination required

✘ Risk of funds lock if keys lost

✘ Higher transaction cost (especially on Ethereum)

Best Practices

- Use 2-of-3 configuration
- Store keys in different physical locations
- Combine hardware wallets (e.g., Ledger)
- Regular audit of smart contracts
- Define clear governance rules

Risks Associated with Multisig

- Collusion among signers
- Key loss (if threshold unmet)
- Smart contract vulnerabilities
- Governance disputes

4. Multi-Signature Wallets

Feature	Single-Key Wallet	Multisig Wallet
Key Required	1	M-of-N
Risk if 1 Key Compromised	Total loss	Partial risk
Enterprise Friendly	No	Yes
Complexity	Low	Medium/High

Courtesy : Himanshi Blog

4. Multi-Signature Wallets

Feature	Multisig Wallet	MPC(Multi-Party Computation) Wallet
Core Idea	Requires multiple private keys to sign a transaction	Private key is never fully created; Signing is done using distributed key shares
Cryptographic Basis	Traditional digital signatures (e.g., ECDSA, Schnorr)	Secure Multi-Party Computation (threshold cryptography)
Key Storage	Each participant holds a complete private key	Each participant holds only a key share

Future of Multisig

- Integration with DeFi
- Cross-chain multisig
- MPC + Multisig hybrid models
- Institutional custody frameworks

Courtesy : Himanshi Blog

5. Securing Wallets: Threats, Recovery Methods, and Best Practices,

Common Wallet Threats

1. Phishing Attacks

- Fake websites mimicking real wallets
- Malicious emails requesting seed phrases

2. Malware & Keyloggers :

- Steal private keys from infected devices

3. SIM Swapping :

- Attackers hijack phone number to bypass 2FA

4. Man-in-the-Middle (MITM) Attacks :

- Intercept wallet communication

5. Social Engineering :

- Impersonation scams

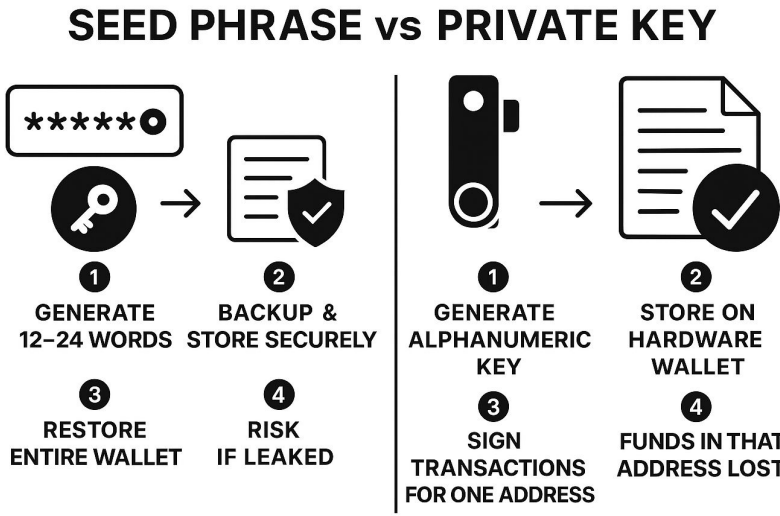
5. Securing Wallets: Threats, Recovery Methods, and Best Practices,

Advanced Threats

- Smart contract exploits
- Clipboard hijacking malware
- Fake browser extensions
- Rug pulls & fake airdrops
- Supply chain attacks

Private Key & Seed Phrase Risks

- Private Key = Full ownership of funds
- Seed Phrase (12–24 words) = Wallet recovery master key
- If exposed → Funds permanently lost
- No central authority for recovery (Decentralized systems)



5. Securing Wallets: Threats, Recovery Methods, and Best Practices,

Wallet Recovery Methods

1. Seed Phrase Recovery
   - Enter 12/24-word recovery phrase
   - Restores wallet on any compatible device
2. Hardware Wallet Backup
   - Restore using recovery seed on new device
3. Multi-Signature Wallet Recovery
   - Requires multiple approvals
   - Used in enterprise setups

Courtesy : Medium

5. Securing Wallets: Threats, Recovery Methods, and Best Practices,

What If Seed Phrase is Lost?

- No password reset option
- No customer care (in decentralized wallets)
- Funds become inaccessible permanently

Major Lost Bitcoin Wallets

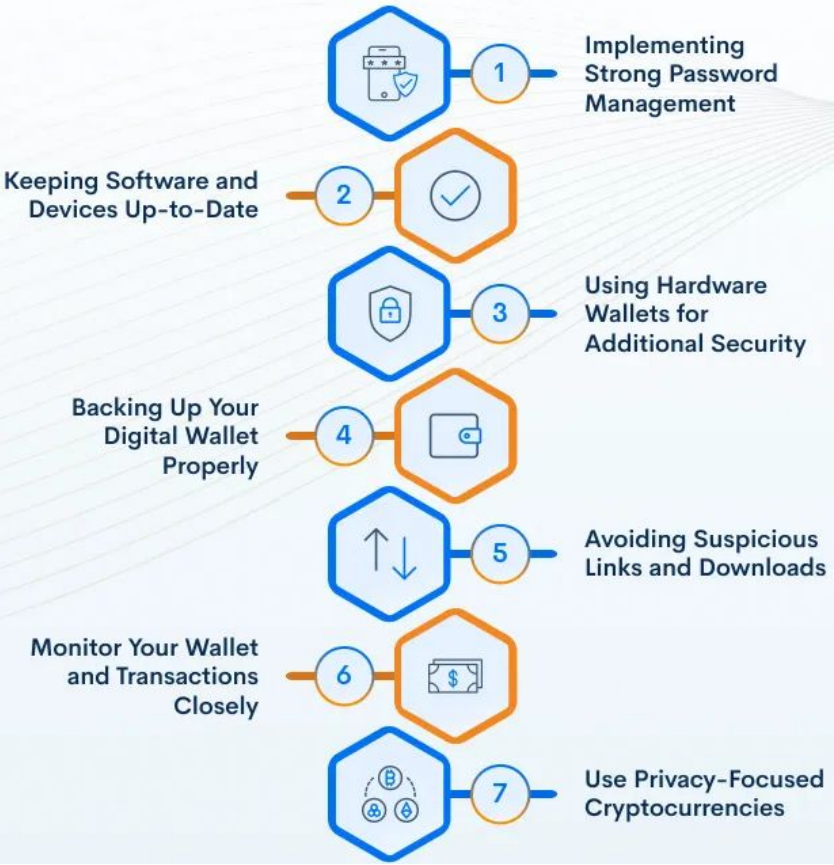
- **Satoshi Nakamoto:** Roughly 1 million BTC mined in 2009–2010, unmoved and likely lost forever.
- **Mt. Gox Hack:** ~79,957 BTC from the 2014 exchange hack.
- **Early Adopters:** Numerous early miners and investors lost access to wallets via damaged hard drives, forgotten passwords, or premature death. CoinTracker +3

Courtesy : Himanshi Blog

5. Securing Wallets: Threats, Recovery Methods, and Best Practices,

Best Security Practices

1. Never Share Seed Phrase (Not even with wallet providers)
2. Use Hardware Wallets for Large Holdings (Example: Ledger)
3. Enable Two-Factor Authentication (2FA)
4. Verify URLs Carefully (Bookmark official sites)
5. Use Dedicated Devices for Crypto



Courtesy : Digital Pay

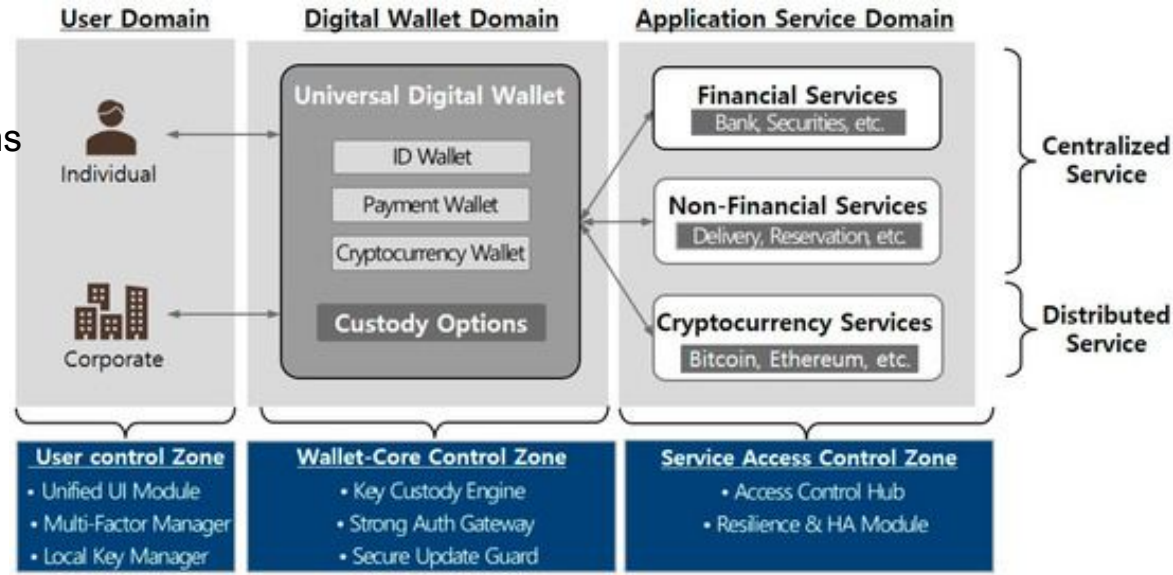
5. Securing Wallets: Threats, Recovery Methods, and Best Practices,

Multi-Layered Security Strategy

- Cold storage for long-term holdings
- Hot wallet for daily transactions
- Multi-signature wallets for organizations
- Regular firmware updates

Institutional Wallet Security

- Role-based access control
- Hardware Security Modules (HSMs)
- Custodial solutions (e.g., exchanges)
- Insurance-backed custody services



Courtesy : MDPI

5. Securing Wallets: Threats, Recovery Methods, and Best Practices,

Real-World Wallet Hacks

- Exchange breaches
- Phishing draining wallets
- SIM swap crypto theft cases

Note : Human error > Technical flaws

Future of Wallet Security

- Biometric authentication
- MPC (Multi-Party Computation) wallets
- Social recovery wallets
- Account abstraction (Ethereum)

Safety Tips

- Wallet security = User responsibility
- Protect private keys & seed phrases
- Use hardware wallets for high-value assets
- Follow layered security approach
- Stay updated on emerging threats

6. Cryptocurrency Exchanges: Types, Trading Mechanisms, and Risks

Cryptocurrency Exchanges

- Platforms where users buy, sell, and trade cryptocurrencies
- Act as intermediaries between buyers and sellers
- Provide liquidity and price discovery
- Core infrastructure for ecosystems like Bitcoin and Ethereum

Role of Exchanges in Crypto Ecosystem

- Market liquidity
- Token listings
- Fiat on/off ramps
- Custody services
- Margin & derivatives trading
- Staking services

6. Cryptocurrency Exchanges: Types, Trading Mechanisms, and Risks

Courtesy : Himanshi Blog

6. Cryptocurrency Exchanges: Types, Trading Mechanisms, and Risks

Types Of Cryptocurrency Exchanges

1. **Centralized Exchanges (CEX)**

- Operated by a company
- Users deposit funds into exchange wallets
- Exchange manages private keys
- High liquidity & fast transactions

Examples:

- Binance
- Coinbase
- Kraken

Advantages:

✔ User-friendly

✔ High trading volume

✔ Customer support

Disadvantages:

✘ Custodial risk

✘ Hacking targets

6. Cryptocurrency Exchanges: Types, Trading Mechanisms, and Risks

Types Of Cryptocurrency Exchanges

2. Decentralized Exchanges (DEX)

- Peer-to-peer trading
- No central authority
- Smart contract-based
- Non-custodial (users control keys)

Examples:

- Uniswap
- PancakeSwap

Advantages:

✔ Greater privacy

✔ No custody risk

✔ Permissionless access

Disadvantages:

✘ Lower liquidity (sometimes)

✘ Smart contract risks

3. Hybrid Exchanges

- Combination of CEX and DEX features
- Faster order execution with decentralized custody models

6. Cryptocurrency Exchanges: Types, Trading Mechanisms, and Risks

Trading Mechanisms

1. **Order Book Model (CEX)**

- Buyers place bids
- Sellers place asks
- Matching engine executes trades
- Price determined by supply & demand
- Used by:
  - **Binance**
  - **Coinbase**

2. **Automated Market Maker (AMM) Model (DEX)**

- No order book
- Liquidity pools
- Smart contracts determine price
- Formula: $x \times y = k$
- Used by: **Uniswap**

6. Cryptocurrency Exchanges: Types, Trading Mechanisms, and Risks

RISKS ASSOCIATED WITH EXCHANGES

1. **Security Risks**
   - Exchange hacks
   - Phishing attacks
   - API key compromise
   - Insider fraud
   - Example: Mt. Gox collapse (2014)
2. **Custodial Risk**
   - Users do not control private keys (CEX)
   - Bankruptcy risk
   - Frozen withdrawals
   - Example: FTX collapse (2022)
3. **Market Risks**
   - High volatility
   - Flash crashes
   - Liquidity risks
   - Manipulation (wash trading, pump & dump
4. **Regulatory Risks**
   - Changing government regulations
   - KYC/AML compliance requirements
   - Exchange bans in certain countries
5. **Smart Contract Risks (DEX)**
   - Code vulnerabilities
   - Impermanent loss
   - Front-running (MEV attacks)

6. Cryptocurrency Exchanges: Types, Trading Mechanisms, and Risks

RISK MITIGATION STRATEGIES

Best Practices for Users

- Use strong passwords & 2FA
- Use hardware wallets for storage
- Avoid keeping large funds on exchanges
- Verify URLs
- Enable withdrawal whitelist

Institutional Risk Controls

- Cold storage reserves
- Proof-of-Reserves audits
- Insurance-backed custody
- Regulatory compliance frameworks

6. Cryptocurrency Exchanges: Types, Trading Mechanisms, and Risks

Feature	Centralized Exchange (CEX)	Decentralized Exchange (DEX)
Control of Funds	Exchange holds your funds	You control your wallet
Custody	Custodial	Non-custodial
KYC Required	Usually Yes	Usually No
Speed	Fast (off-chain matching)	Slower (on-chain transactions)
Fees	Trading fees	Gas fees + protocol fee
Security Risk	Risk of hacks / company failure	Smart contract risk
Privacy	Lower (identity verification)	Higher
Ease of Use	Beginner-friendly	Requires wallet knowledge
Token Listings	Limited (approved tokens)	Open listing (any token)
Customer Support	Yes	No central support

Courtesy : Himanshi Blog

7. Regulatory and Legal Frameworks for Cryptocurrencies Globally

- Cryptocurrencies operate in a borderless digital environment
- Regulation varies significantly across countries
- Key concerns:
  - Financial stability
  - Investor protection
  - Anti-Money Laundering (AML)
  - Counter-Terrorism Financing (CTF)
  - Taxation
- Regulatory bodies are adapting to blockchain innovations

Why Regulation is Needed?

- Prevent fraud and scams
- Protect retail investors
- Prevent money laundering
- Ensure financial stability
- Enable innovation within legal boundaries

7. Regulatory and Legal Frameworks for Cryptocurrencies Globally

GLOBAL REGULATORY LANDSCAPE

United States

Regulatory approach: Fragmented but evolving

Key regulators:

- Securities and Exchange Commission (Securities regulation)
- Commodity Futures Trading Commission (Commodity oversight)
- Financial Crimes Enforcement Network (AML compliance)

Highlights:

- Bitcoin often treated as a commodity
- ICOs may be classified as securities
- Strong enforcement actions

7. Regulatory and Legal Frameworks for Cryptocurrencies Globally

GLOBAL REGULATORY LANDSCAPE

India

Regulatory status: Cautious but not banned

Key authorities:

- Reserve Bank of India
- Securities and Exchange Board of India

Key points:

- 30% tax on crypto gains
- 1% TDS on transactions
- AML compliance under PMLA
- No formal classification yet as legal tender

European Union

Regulatory approach: Unified framework

Key development:

- Markets in Crypto-Assets Regulation (MiCA)

Features:

- Licensing requirements
- Stablecoin regulation
- Consumer protection rules
- Passporting across EU member states

7. Regulatory and Legal Frameworks for Cryptocurrencies Globally

GLOBAL REGULATORY LANDSCAPE

China

Regulatory stance: Strict prohibition

- Crypto trading banned
- Mining activities restricted
- Promotion of state-backed digital currency

Digital Yuan by: **People's Bank of China**

Japan

Regulatory stance: Crypto-friendly but regulated

Regulator: Financial Services Agency

Key features:

- Exchanges must register
- Strict AML/KYC rules
- Recognizes crypto as legal property

7. Regulatory and Legal Frameworks for Cryptocurrencies Globally

LEGAL CLASSIFICATION OF CRYPTO ASSETS

Cryptocurrencies may be treated as:

- Commodity (e.g., Bitcoin in US)
- Security (certain tokens/ICOs)
- Property (for tax purposes)
- Payment instrument
- Virtual digital asset (India)

7. Regulatory and Legal Frameworks for Cryptocurrencies Globally

AML / KYC COMPLIANCE

FATF Guidelines

- Global standards by: Financial Action Task Force (FATF) Guideline
- Key Requirements:
  - Travel Rule
  - Customer identification
  - Reporting suspicious transactions

Taxation Policies

Common tax treatments:

- Capital gains tax
- Income tax on trading profits
- VAT in some jurisdictions
- TDS (India model)

7. Regulatory and Legal Frameworks for Cryptocurrencies Globally

LEGAL RISKS & CHALLENGES

Regulatory Challenges

- Cross-border jurisdiction issues
- Decentralized governance
- Privacy coins
- DeFi regulation difficulty
- NFT legal ambiguity

DeFi and Regulatory Dilemma

- No central intermediary
- Smart contract-based operations
- Difficult enforcement
- DAO liability questions

Example ecosystem: Ethereum

CBDCs vs Cryptocurrencies

Central Bank Digital Currencies (CBDCs):

Examples:

- Digital Yuan (China)
- Digital Euro (EU under study)
- Pilot projects in India

Difference:

- Centralized control vs decentralized crypto

7. Regulatory and Legal Frameworks for Cryptocurrencies Globally

Emerging Trends

- Global harmonization efforts
- Stablecoin-specific regulations
- Licensing of exchanges
- Institutional adoption
- Real-world asset tokenization

Key Takeaways

- No universal global framework yet
- Regulation ranges from supportive to prohibitive
- Compliance (AML/KYC) is becoming mandatory worldwide
- Legal clarity influences adoption
- Regulatory clarity is essential for sustainable growth
- Balance required between innovation and investor protection
- Global cooperation will shape the future of crypto law

Courtesy : Himanshi Blog

7. Regulatory and Legal Frameworks for Cryptocurrencies Globally

Region	Regulatory Stance	Legal Status	Taxation
USA	Regulated	Commodity/Security	Capital Gains
EU	MiCA Framework	Legal	Varies
India	Taxed but unregulated	VDA	30%
China	Banned	Illegal	N/A
Japan	Regulated	Legal property	Taxable

Courtesy : Himanshi Blog